

Clinical Risk Assessment Report: Patient 67

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.50x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: -0.3 [Bottom 24%] (+1.3%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.7 [Avg (28th %ile)] (+1.2%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: -0.58 [Avg (26th %ile)] (-3.6%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Patient Age (Age)

Value: 70.0 [Top 25%] (-1.7%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT)

Value: 1.08 [Top 15%] (-1.2%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

Value: 0.14 [Avg (43th %ile)] (-1.0%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: -0.53 [Avg (30th %ile)] (-0.7%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.
- Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET)

Value: -0.49 [Avg (38th %ile)] (-0.5%)

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Machine Learning Feature Attribution (SHAP Decision Path)

